

TATHAGATA BANDYOPADHYAY

Munich, Germany

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Profile Summary

Experienced software developer with a passion for research, particularly in the fields of Deep Learning and AI. Seeking a research role related to Generative AI, encompassing either of Large Language Models (LLMs), 2D/3D Computer Vision, Audio Processing or multi-modal AI using Auto-encoders, GANs, or Diffusion Models. Open for collaboration on research projects in the broad domains of Deep Learning and Generative AI.

Education

Technical University of Munich

Oct. 2019 – Mar. 2024

M.Sc. in Informatics | Grade: 1.4 (in German Scale) | Passed with Distinction

Munich, Germany

KIIT University

Aug. 2013 – May 2017

B.Tech. in Computer Science | Grade: 1.3 (in German Scale) | Founder's All-rounder Gold Medalist

Bhubaneswar, India

Experience

University of Tübingen (AI Center)

Jun. 2025 – Present

Research Intern at Autonomous Learning Group

Tübingen, Germany

- Sparse feed-forward 3D Gaussian Splatting (3DGS) for multi-view 3D reconstruction.
- Patch-particle based efficient hybrid representation for neural spatio-temporal world modeling.

Max Planck Institute for Intelligent Systems

Dec. 2024 – May 2025

Research Intern at Optics and Sensing Lab

Tübingen, Germany

- Video understanding for analyzing non-verbal cues of human communication using computer vision
- Semantic latent representation learning for diverse human attributes like pose, style, skin-tone etc.

Siemens (Technology)

Mar. 2023 – Sep. 2024

Working Student Researcher

Munich, Germany

- Developed an Intuitive Natural Language Query system for Knowledge Graphs using LLMs.
- Implemented Retrieval Augmented Generation (RAG) on Knowledge Graphs using graph traversal
- Filed 3 Invention Disclosures (Patents) and published 2 Research Papers.

Technical University of Munich

Apr. 2021 – Sep. 2022

Teaching Assistant (TA) for Introduction to Deep Learning (I2DL IN2346)

Munich, Germany

- Designed hands-on exercises for Deep Learning course using PyTorch, NumPy and Python.
- Taught Deep Learning Exercises to more than 1000 international students of TUM.
- Prepared examination questions for Deep Learning and graded student answer papers.

Dell Technologies

Jul. 2017 – Sep. 2019

Software Engineer I

Bangalore, India

- Developed an Intelligent Price Recommendation Engine with Statistical Modelling of historical price data.
- Utilized Web Scrapping using Python, Pandas, NumPy and BeautifulSoup to collect and analyse market price.
- Applied Test Driven Development (TDD) and CI-CD pipelines to improve code quality

Publications

- **Tathagata Bandyopadhyay**. Spectron: Target speaker extraction using conditional transformer with adversarial refinement. *arXiv preprint arXiv:2409.01352*, 2024
- Nilay Tufek-Oezkaya, Burak Yigit Uslu, Valentin Philipp Just, **Tathagata Bandyopadhyay**, Aparna Saissre Thuluva, Marta Sabou, and Allan Hanbury. A new retrieval-augmented generation (rag) approach for querying and constructing large-scale knowledge graphs. *semantic-web-journal*, 2024
- Nilay Tufek, Aparna Saissre, Valentin Philipp Just, Fajar J. Ekputra, **Tathagata Bandyopadhyay**, Marta Sabou, and Allan Hanbury. Validating semantic artifacts with large language models. In *Proceedings of The Semantic Web – 21st International Conference, ESWC 2024*. Springer, ESWC 2024, 2024
- **Tathagata Bandyopadhyay**, Jan Kubicek, Marek Penhaker, Juraj Timkovic, David Oczka, and Ondrej Krejcar. A semi-supervised learning approach for automatic segmentation of retinal lesions using surf blob detector and locally adaptive binarization. In *Proceedings of the 11th ASIAN Conference on Intelligent Information and Database Systems (ACIIDS 2019)*, pages 311–323. Springer, LNCS(LNAI), volume 11432, 2019

- **Tathagata Bandyopadhyay**, Shyamali Mitra, Sreetama Mitra, Luis Miguel Rato, Nibaran Das, and Mrinal Kanti Naskar. A fast algorithm for automatic segmentation of pancreas histological images for glucose intolerance identification. In *Recent Developments in Machine Learning and Data Analytics (IC3 2018)*, pages 307–315. Springer, AISC, volume 740, 2018
- Rajdeep Chatterjee, **Tathagata Bandyopadhyay**, Debarshi Kumar Sanyal, and Dibyajyoti Guha. Comparative analysis of feature extraction techniques in motor-imagery eeg signal classification. In *Proceedings of First International Conference on Smart Systems, Innovations and Computing*, pages 73–83. Springer, SIST, volume 79, 2018
- **Tathagata Bandyopadhyay**, Sreetama Mitra, Shyamali Mitra, Luis Miguel Rato, and Nibaran Das. Analysis of pancreas histological images for glucose intolerance identification using wavelet decomposition. In *Proceedings of the 5th International Conference on Frontiers in Intelligent Computing: Theory and Applications*, pages 653–661. Springer, AISC, volume 515, 2017
- Rajdeep Chatterjee, **Tathagata Bandyopadhyay**, Debarshi Kumar Sanyal, and Dibyajyoti Guha. Dimensionality reduction for eeg signal using fuzzy discernibility matrix. In *Proceedings of the 10th International Conference on Human System Interaction (HSI)*, pages 131–136. IEEE, 2017
- Rajdeep Chatterjee, **Tathagata Bandyopadhyay**, and Debarshi Kumar Sanyal. Effects of wavelets on quality of features in motor-imagery eeg signal classification. In *Proceedings of International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET)*, pages 1346–1350. IEEE, 2016
- Rajdeep Chatterjee and **Tathagata Bandyopadhyay**. Eeg based motor imagery classification using svm and mlp. In *Proceedings of 2nd International Conference on Computational Intelligence and Networks (CINE)*, pages 84–89. IEEE, 2016

Patents

- An LLM based method for contextual dynamic entity extraction from a Natural Language Sentence and linking them to a structured text source. (filed)
- A graph traversal method for contextual sub-graph extraction from a knowledge graph for natural language based interactive question answering. (Submitted)
- Simplifying the generation of formal queries from natural language OPC UA queries using Large Language Models. (filed)

Projects

Neural Parametric 3D Head Models with 2D GANs | *Python, PyTorch, PyVista, Meshlab, Wandb* **Dec. 2023**

- Developed a neural parametric head model to reconstruct 3D head geometry from partial point clouds or depth scans.
- Utilized neural network to predict 3D signed distance field as an implicit 3D head representation.
- Implemented differentiable volumetric neural rendering to compute 2D GAN objectives from multi-view normal maps.

Conditional Speech Separation with Transformers | *Python, PyTorch, Librosa, VScode, Wandb* **Mar. 2022**

- Developed a target speaker extraction system from monaural mixed-speech audio using transformer neural networks.
- Utilized LSTM and 1D CNN for encoding the inputs and generative adversarial objectives to further refine the outputs.
- Used Python and PyTorch for neural networks, VSCode for IDE and Librosa for audio mixing and transformations.

Semi-Supervised Representation Learning using Group Loss | *Python, PyTorch, PyCharm* **Mar. 2021**

- Designed a semi-supervised image classification algorithm using group loss to consider intra-batch embedding similarity.
- Improved image label guessing and label propagation using group loss based deep metric learning.
- Used Python and PyTorch for implementation and used MixMatch paper from Google Research as baseline.

Technical Skills

Programming Languages: Python, C#, C++, Java, C, HTML/CSS, JavaScript

Technologies/Frameworks: PyTorch, LangChain, FastAPI, FLASK, Git, .Net, Streamlit

Developer Tools: PyCharm, VS Code, Visual Studio, Docker, MATLAB, Wandb

Topics: Deep Learning, Computer Vision, 3D Reconstruction, Generative AI, LLMs, REST APIs